

# Limit theorems

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## The Central Limit Theorem

*Remark.* The sum of many independent random variables, none of which dominates the others, has a distribution that is approximately bell-shaped, whatever the distributions of the individual variables. Theorem 0.1 makes this precise.

**Theorem 0.1** (Central Limit Theorem). *Let  $X_1, X_2, \dots$  be IID random variables with mean  $\mu$  and finite variance  $\sigma^2 > 0$ , and let  $S_n \stackrel{\text{def}}{=} \sum_{i=1}^n X_i$ . Then for every real number  $z$ :*

$$\lim_{n \rightarrow \infty} \Pr\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq z\right) = \Phi(z)$$

where  $\Phi(z) \stackrel{\text{def}}{=} \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du$  is the CDF of the *standard normal distribution*  $N(0, 1)$ .

*Remark.* This version is the Lindeberg–Lévy CLT; its proof is beyond these notes’ scope (Billingsley 1995, Theorem 27.1). Other versions relax the IID assumption, which is why the informal statement asks only that no summand dominate.

**Corollary 0.1** (Mean and variance of a sum of IID random variables). *Let  $X_1, \dots, X_n$  be IID random variables, each discrete or continuous, with mean  $\mu$  and finite variance  $\sigma^2$ , and let  $S_n \stackrel{\text{def}}{=} \sum_{i=1}^n X_i$ . Then:*

$$\begin{aligned} \mathbb{E}[S_n] &= n\mu \\ \text{Var}(S_n) &= n\sigma^2 \end{aligned}$$

### **i** Proof

*Proof.* For the mean, apply [linearity of expectation](#) once per added summand:

$$\begin{aligned} \mathbb{E}[S_n] &= \sum_{i=1}^n \mathbb{E}[X_i] \quad (\text{linearity of expectation, applied } n-1 \text{ times}) \\ &= n\mu \quad (\text{each } X_i \text{ has mean } \mu) \end{aligned}$$

For the variance, apply the [variance of a linear combination](#) with every  $a_i = 1$ . For  $i \neq j$ ,  $X_i$  and  $X_j$  are independent (take  $A_k = \mathbb{R}$  for every other  $k$  in the [definition of independence](#)), so  $\text{Cov}(X_i, X_j) = 0$  for [independent summands](#); and  $\text{Cov}(X_i, X_i) = \text{Var}(X_i)$  ([covariance of a variable with itself](#)):

$$\begin{aligned} \text{Var}(S_n) &= \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j) && (\text{variance of a linear combination, all } a_i = 1) \\ &= \sum_{i=1}^n \text{Cov}(X_i, X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j) && (\text{split off the terms with } i = j) \\ &= \sum_{i=1}^n \text{Var}(X_i) + 0 && (\text{covariance with itself; independent summands}) \\ &= n\sigma^2 && (\text{each } X_i \text{ has variance } \sigma^2) \end{aligned}$$

□

*Remark.* In practice, Theorem [0.1](#) justifies approximating  $S_n$  by a normal distribution with mean  $n\mu$  and variance  $n\sigma^2$ , which by Corollary [0.1](#) are exactly the mean and variance of  $S_n$ .

**Example 0.1** (The sum of five dice). A single fair die roll has the discrete uniform distribution on  $\{1, \dots, 6\}$ , which is flat, not bell-shaped (Figure [1](#)). Its mean is  $\mu = 3.5$ , and its variance is  $\sigma^2 = \sum_{x=1}^6 (x - 3.5)^2 / 6 = 35/12$ .

```

dice_sum_pmf <- function(n_dice) {
  totals <- rowSums(expand.grid(rep(list(1:6), n_dice)))
  probs <- prop.table(table(totals))
  data.frame(total = as.numeric(names(probs)), p = as.vector(probs))
}

dice_plot <-
  ggplot2::ggplot(mapping = ggplot2::aes(x = total, y = p)) +
  ggplot2::xlab("sum of dice (x)") +
  ggplot2::ylab("Probability of outcome, Pr(X=x)") +
  ggplot2::expand_limits(y = 0)

dice_plot + ggplot2::geom_col(data = dice_sum_pmf(1))

```



Figure 1: Distribution of the outcome of one die

The sum of five independent rolls,  $S_5$ , is already close to bell-shaped (Figure 2).

```

pmf_five_dice <- dice_sum_pmf(5)
dice_plot + ggplot2::geom_col(data = pmf_five_dice)

```

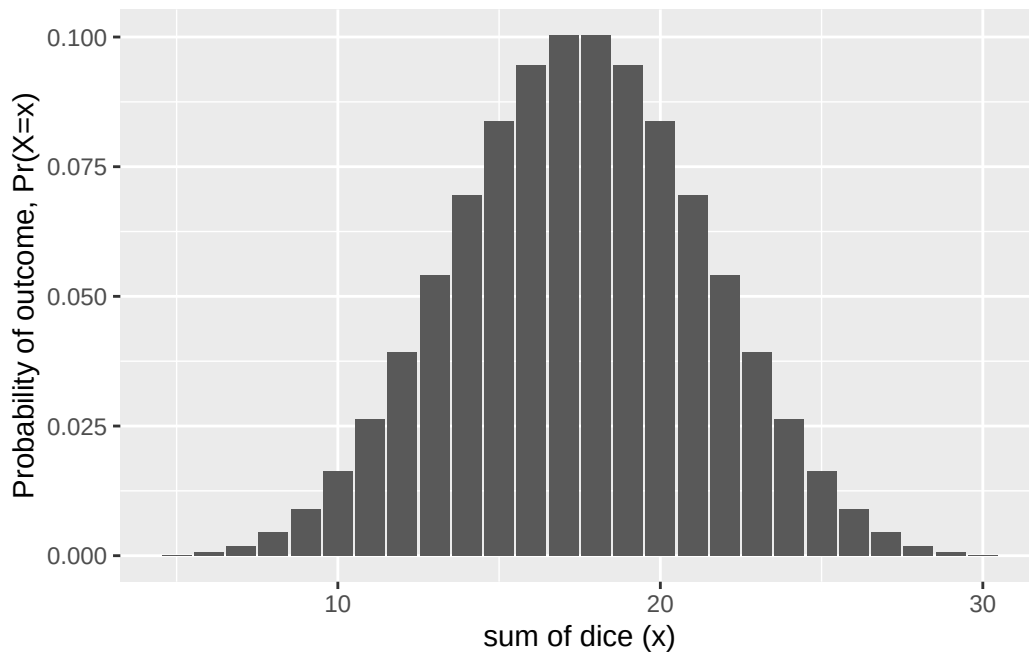


Figure 2: Distribution of the sum of five dice

For example, the exact probability that five dice total at most 15 is  $\Pr(S_5 \leq 15) = 0.3052$ . The normal approximation that Theorem 0.1 suggests, with mean  $5 \cdot 3.5 = 17.5$  and variance  $5 \cdot 35/12 \approx 14.58$ , evaluated at 15.5 to account for  $S_5$  taking only integer values, gives  $\Phi((15.5 - 17.5)/\sqrt{14.58}) \approx 0.3002$ .

## References

Billingsley, Patrick. 1995. *Probability and Measure*. 3rd ed. Wiley Series in Probability and Mathematical Statistics. Wiley.